

4 Results

4.1 Cohort and downstream models

The manifest contained 1,686 unique patients: 1,180 training, 253 validation, and 253 locked-test patients. All results below are from development data; the locked test remained sealed. Five-fold patient-level out-of-fold (OOF) prostate segmentation yielded 1,016 eligible training predictions, with mean Dice 0.912, IoU 0.842, ASD 1.172, and 95HD 3.305. The selected downstream model achieved Dice 0.917 (95% bootstrap CI 0.907-0.924), IoU 0.851 (0.839-0.861), ASD 1.135 (1.041-1.265), and 95HD 3.132 (2.897-3.420) in 217 eligible validation patients.

OOF predictions generated 180 reject, 459 caution, and 377 accept training labels; the validation distribution was 22, 93, and 102, respectively. The independent diagnostic models achieved AUC 0.766 (0.693-0.837) for PI-RADS ≥ 4 and 0.710 (0.630-0.781) for Gleason grade group ≥ 2 in 198 validation patients. Their PR-AUC values were 0.780 and 0.706.

4.2 Ternary IQA performance

The prostate-segmentation-derived IQA checkpoint selected at epoch 8 achieved accuracy 0.558 (95% bootstrap CI 0.488-0.622), macro F1 0.499 (0.420-0.572), quadratic-weighted kappa 0.359 (0.231-0.480), and ordinal MAE 0.498 (0.419-0.581). Recall was 0.409 for reject, 0.441 for caution, and 0.696 for accept. Despite hard-class overlap, mean expected ordinal score increased monotonically from 0.822 for reject to 1.092 for caution and 1.400 for accept.

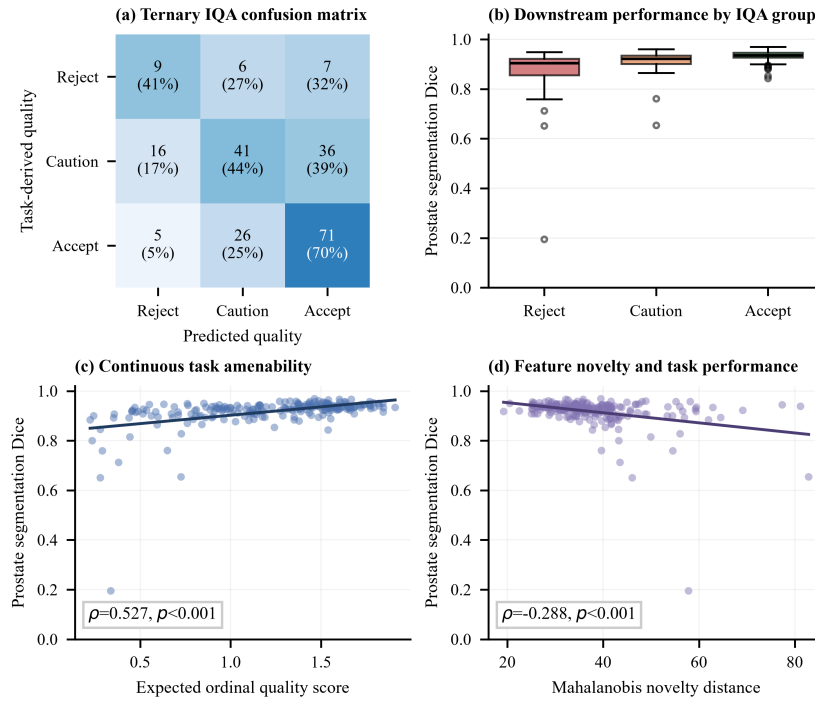


Fig. 1. Development-validation behavior of the prostate-segmentation-derived IQA model ($n=217$). **(a)** Counts and row-normalized confusion matrix. **(b)** Dice by predicted quality. **(c)** Expected ordinal score against Dice. **(d)** feature-space novelty against Dice.

4.3 Association with downstream performance

Predicted groups showed a monotonic performance gradient (Table 1). Mean Dice rose from 0.858 in reject to 0.913 in caution and 0.934 in accept; IoU increased and both surface-distance metrics decreased. Expected quality score correlated with Dice and IoU (Spearman $r=0.527$, 95% bootstrap CI 0.422-0.619, $p<0.001$), ASD ($r_s=-0.277$, $p<0.001$), and 95HD ($r_s=-0.219$, $p=0.001$).

Table 1. Prostate segmentation performance by predicted IQA group. Values are mean (standard deviation).

Group	<i>n</i>	Dice	IoU	ASD	95HD
Reject	30	0.858 (0.144)	0.772 (0.162)	1.699 (1.954)	4.451 (4.271)
Caution	73	0.913 (0.042)	0.842 (0.062)	1.108 (0.474)	3.009 (1.274)
Accept	114	0.934 (0.021)	0.877 (0.035)	1.005 (0.323)	2.864 (1.048)

Softmax confidence was only weakly associated with Dice ($r_s=0.162$, $p=0.017$), and entropy was not ($r_s=-0.113$, $p=0.096$). Mahalanobis novelty distance was more informative: it correlated negatively with Dice ($r=-0.288$, 95% CI -0.415 to -0.158, $p<0.001$) and positively with ASD ($r=0.260$, $p<0.001$) and 95HD ($r=0.219$, $p=0.001$). Mean novelty distance was 47.31, 39.30, and 34.63 in predicted reject, caution, and accept.

4.4 Cross-task transfer

Prostate-segmentation-derived expected quality was not associated with PI-RADS correctness ($r_s=-0.026$, $p=0.716$) or Gleason correctness ($r_s=0.058$, $p=0.418$). Thus, scans estimated to be amenable to gland segmentation were not necessarily amenable to diagnostic classification.

5 Discussion

5.1 Interpretation

These development results show that prostate MRI quality can be learned from downstream success without direct visual-quality labels. The main evidence is the monotonic separation of segmentation performance: the absolute mean-Dice difference between predicted reject and accept was 0.076, with concordant changes in overlap and boundary metrics. Moderate hard-class performance does not erase this signal. Task performance is continuous, while reject/caution/accept labels are thresholded operational categories; cases close to a Dice or 95HD boundary are expected to move between adjacent classes. The expected ordinal score may therefore be more informative than a forced three-way decision, while the categories retain a useful triage interface.

5.2 Task specificity and reliability

The lack of transfer to PI-RADS and Gleason correctness supports, rather than contradicts, the task-amenability hypothesis. Gland segmentation depends chiefly on boundary visibility and anatomical contrast; diagnostic classification additionally depends on focal lesion signal, texture, zonal context, and clinical-label quality. A single universal quality target would conceal these differences. A defensible extension is a shared encoder with separate anatomical, lesion, PI-RADS, and Gleason quality heads.

Novelty was more closely related to segmentation failure than entropy or confidence, consistent with the distinction between ambiguity among known classes and departure from the training distribution. Combining ordinal quality with novelty could therefore provide a safety layer in which an apparently acceptable but unfamiliar scan is escalated to caution. Any novelty cutoff must, however, be tuned on development data and frozen before locked testing.

5.3 Limitations

The validation split contributed to checkpoint and threshold selection; these results are not an unbiased estimate of final generalization. The 253-patient locked test set must be evaluated once with frozen weights, preprocessing, label thresholds, and operating points, followed by external-site validation. Task-derived labels also inherit downstream-model errors and threshold choices and therefore describe model amenability, not a universal radiological-quality construct. The small reject class limited reject recall and widened uncertainty.

Only cases with complete task-specific inputs and targets entered each analysis. The prostate mask was excluded from primary IQA image inputs to avoid direct target leakage, but segmentation supervision can still reflect annotation variability and anatomical difficulty as well as image quality. OOF lesion-segmentation-derived IQA was not available in the downloaded results and should be reported separately when complete. Correlation tests remain exploratory and unadjusted for multiplicity; the bootstrap procedure must be prespecified before locked testing.

6 Conclusion

Task-derived ternary prostate MRI IQA was technically feasible and learned an ordinal signal that stratified prostate-segmentation Dice from 0.858 in predicted reject to 0.934 in accept. Novelty added complementary reliability information, while non-transfer to PI-RADS and Gleason reinforced that quality is task specific. The current evidence supports a proof of concept for anatomical task amenability; locked-test, external, and lesion-task validation are required before clinical interpretation.